

Qatar-1b

R-2342

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_111122 · created 2026-08-19T00:30:41+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455887.61701 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.69 +/- 2.1 [%]
Orbital Inclination (inc)	88.79 +/- 2.92
Airmass coefficient 1 (a1)	3.4 +/- 2.29
Airmass coefficient 2 (a2)	-0.9 +/- 0.68
Scatter in the residuals of the lightcurve fit is	75.92 %
Best Comparison Star	#1 - [184, 426]
Optimal Aperture	3.66
Optimal Annulus	8.85
Transit Duration (day)	0.083 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.055 vs 0.1463 (deviation 0.82 σ)
Duration fit vs published	0.083 d vs 0.0692 d (deviation 1.95 σ)
Mid-time O-C	-0.2 min
Depth SNR	3.5
Residual scatter	75.92 %

Light curve

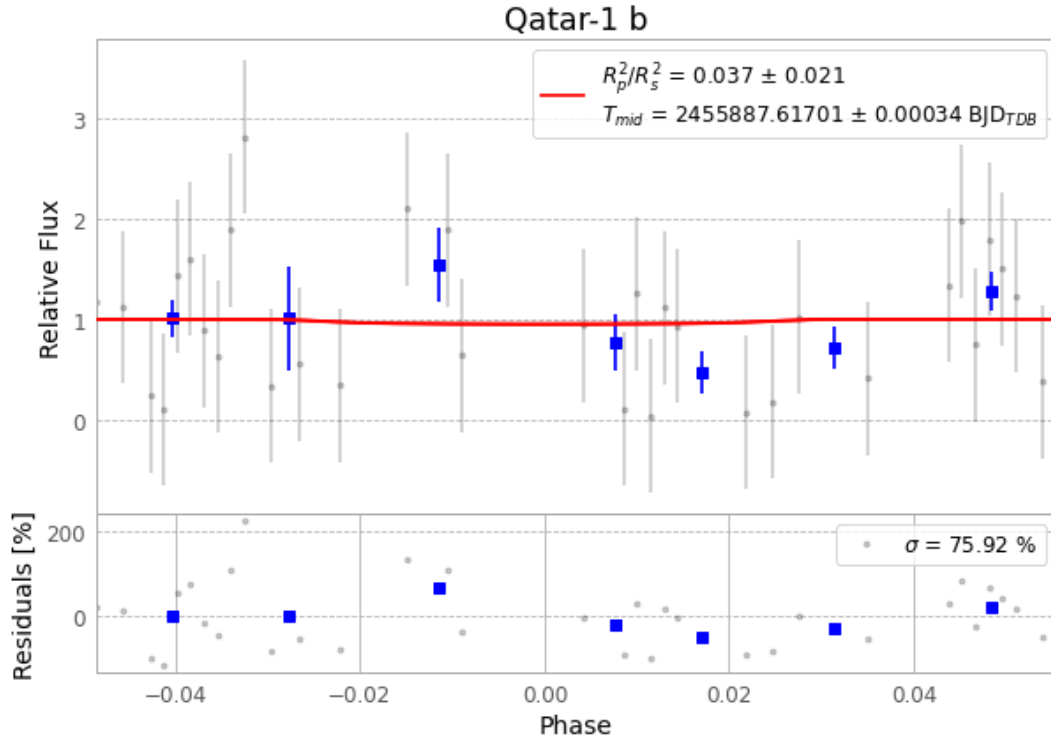


Figure 1 — FinalLightCurve_Qatar-1 b_2011-11-21.png (SHA-256 0debe8c6436dbb75...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 16039430480cd33f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

72 FITS frames (47 MB), Qatar-1111122010612.FITS → Qatar-1111122043912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-111122.log (6008995d8129...), FinalLightCurve_Qatar-1 b_2011-11-21.png (0debe8c6436d...), FinalLightCurve_Qatar-1 b_2011-11-21.csv (8675363904c4...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:30Z.